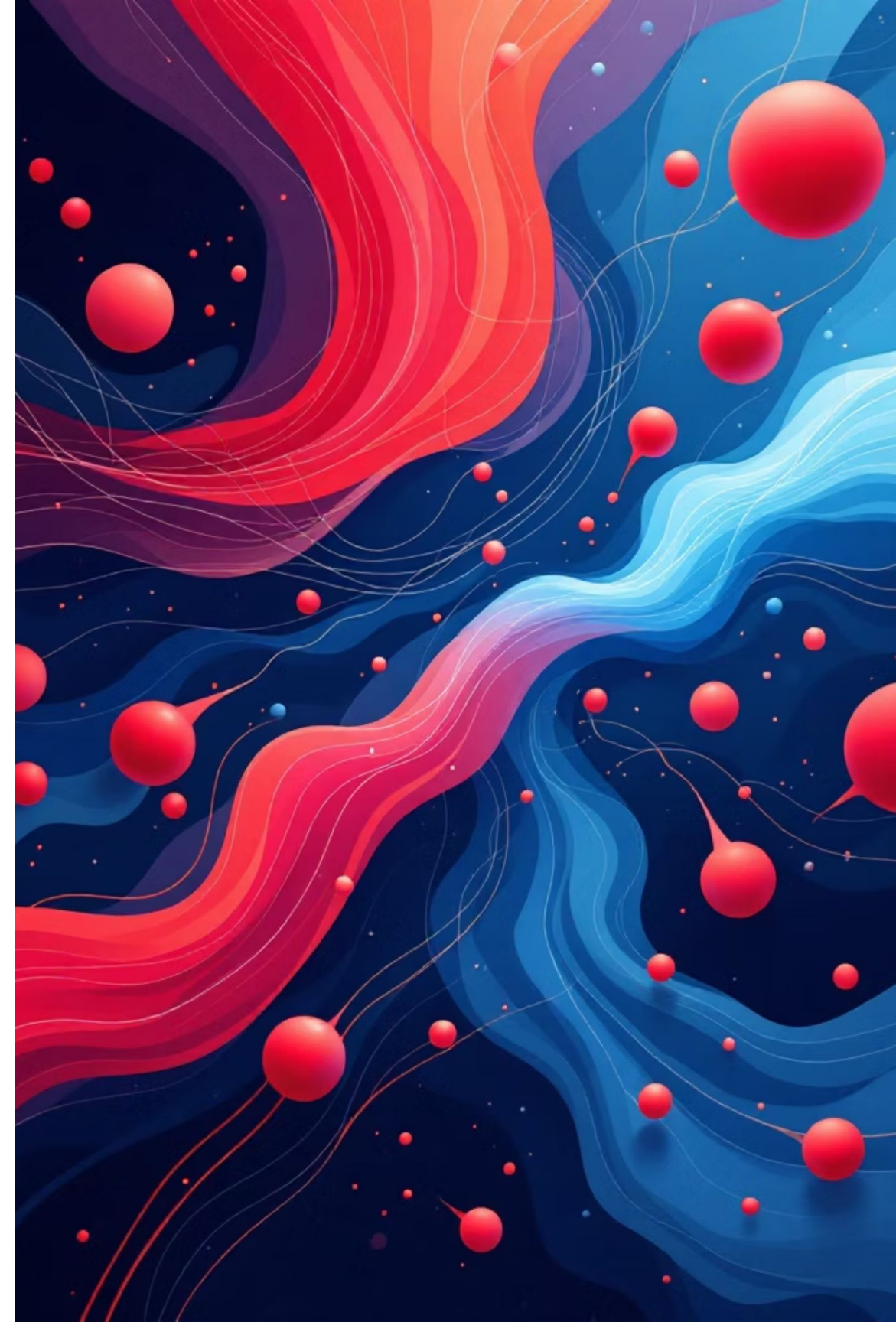


# AI for Communication and Marketing

Apr 01, 2026 - Lab lecture #4: Cluster Analysis



# Beyond Demographics: What Is Cluster Analysis?

## The Old Way

Traditional segmentation divides audiences by age, gender, or geography — blunt instruments that miss behavioral nuance and fail to predict real purchasing decisions.

## The Cluster Analysis Advantage

Cluster Analysis is an **unsupervised machine learning technique** that groups customers by *actual similarity* — behavior, preferences, engagement patterns — revealing segments you never knew existed.



# The Core Principle: Similarity at Scale

Two mathematical properties define every good clustering solution — and every actionable marketing segment.

## Internal Homogeneity

Customers **within** the same cluster are as similar to each other as possible — sharing behaviors, values, and responses to messaging.



## External Heterogeneity

Customers **across** different clusters are as distinct as possible — ensuring each segment responds differently and warrants its own strategy.

- ❏ Think of it this way: **tight clusters, wide gaps**. The more distinct the groups, the more targeted — and effective — your communication can be.



# Step 1 — Data Preparation: Choosing the Right Variables



## Garbage In, Garbage Out

Your clusters are only as meaningful as the variables you feed the algorithm. Prioritize **behavioral and attitudinal KPIs** over raw demographics.

### Behavioral

Purchase frequency, channel usage, content engagement, session depth

### Attitudinal

Survey scores, NPS, brand sentiment, stated preferences

### Transactional

Average order value, RFM scores (Recency, Frequency, Monetary)

# Step 2 — Choosing the Right Algorithm

Two algorithms dominate marketing applications. Each has a distinct logic — choose based on your data size and analytical goal.



## K-Means Clustering

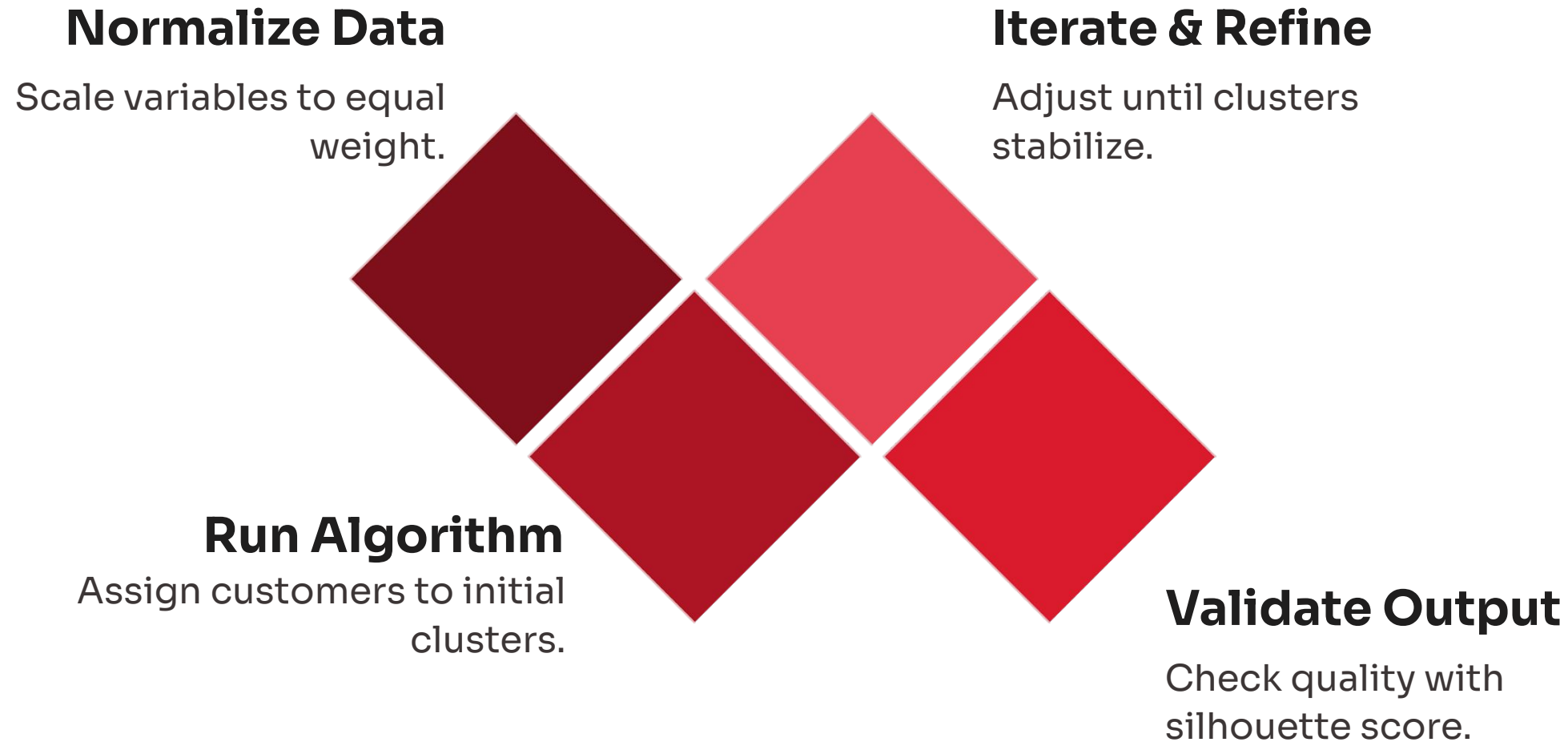
**Best for large datasets.** You specify the number of clusters (k) upfront. The algorithm iteratively assigns each customer to the nearest centroid. Fast, scalable, ideal for CRM and campaign targeting.



## Hierarchical Clustering

**Best for exploration.** Builds a dendrogram (tree) of nested clusters — no need to pre-specify k. Ideal for smaller datasets and discovering natural audience structures before scaling.

# Step 3 — The Clustering Process



Normalization is critical — without it, a variable like "annual spend (\$10,000)" will dominate a variable like "NPS score (0–10)," distorting your segments. Validation metrics like the **silhouette score** confirm your clusters are statistically meaningful before you act on them.





# From Numbers to Personas: Naming Your Clusters

Raw cluster IDs ("Cluster 1, 2, 3") mean nothing to a marketing team. The real value comes from translating statistical groupings into **vivid, actionable personas**.



## Analyze the Profile

Examine the mean values of each variable per cluster. What behaviors define this group?



## Assign a Name

Give each cluster a memorable label:

*"The Loyal Advocates,"*

*"Price-Sensitive Browsers,"*

*"High-Value Dormants."*



## Write the Narrative

Draft a one-paragraph persona: who they are, what they want, what message will resonate.

# Hands-on Session: Your Cluster analysis challenge

Dataset

German Credit Risk

In this dataset, each entry represents a person who takes a credit by a bank. Each person is classified as good or bad credit risks according to the set of attributes.

The provided dataset includes 1.000 observations and 10 attributes, which assume the meaning given in Table 1.

| Attribute ID | Attribute Name   | Attribute Type | Attribute Description                                                                                                                 |
|--------------|------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1            | ID               | Numeric        | Unique Identifier                                                                                                                     |
| 2            | Age              | Numeric        | Age of the customer                                                                                                                   |
| 3            | Sex              | String         | Gender of the customer (male, female)                                                                                                 |
| 4            | Job              | Numeric        | Job type of the customer (0- unskillend and non-resident, 1 - unskilled and resident, 2- skilled, 3 - highly skilled)                 |
| 5            | Housing          | String         | House ownership type (own, rent, free)                                                                                                |
| 6            | Saving accounts  | String         | Saving accounts amount availability (little, moderate, quite rich, rich)                                                              |
| 7            | Checking account | Numeric        | Checking account amount in Deutsch Mark (DM)                                                                                          |
| 8            | Credit Amount    | Numeric        | Credit amount requested in Deutsch Mark (DM)                                                                                          |
| 9            | Duration         | Numeric        | Credit duration in months                                                                                                             |
| 10           | Purpose          | String         | Reason of the credit request (car, furniture/equipment, radio/TV, domestic appliances, repairs, education, business, vacation/others) |



# Hands-on Session: Your Cluster analysis challenge

01

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## Dataset Audit & Quality

Begin with a comprehensive audit of `german_credit_data.csv` to understand the structure and key characteristics. Identify and address missing values and outliers.

03

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## Pre-processing and Clustering

Prepare the data for Cluster Analysis. Encode categorical variables and normalize data. Perform both a K-Means and a Hierarchical Cluster Analysis: use the Elbow Method and Silhouette Score to determine the optimal number of clusters and compare the results.

02

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## EDA

Perform an Exploratory Data Analysis (EDA) including a correlation analysis for the numerical variables.

04

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## Insights and interpretation

Analyze the resulting clusters. Provide a brief summary describing the profile of the typical customer in each group.